# To-Do List App

Manage daily tasks with add, edit, delete and completion controls, a RecyclerView list, status filters and durable local SQLite storage.

## Requirement coverage

**Task CRUD** - Each task has a required title, optional details and a stable ID. SQLite stores all fields. A custom ListAdapter binds each item and its actions.

**Completion and filtering** - A checkbox persists the done flag and adds a strike-through title. All/Open/Done filters and literal text search can be combined. The count reports completed versus total tasks.

**Lifecycle and feedback** - Invalid titles show field errors. Delete requires confirmation. Background database work avoids blocking the UI; ViewModel retains in-flight editor saves.

## Run and review

Open the source folder in Android Studio, install SDK 35 / Build Tools 35.0.0 and use JDK 17 or 21. Sync Gradle and run the todo module on Android 8.0 / API 26 or later.

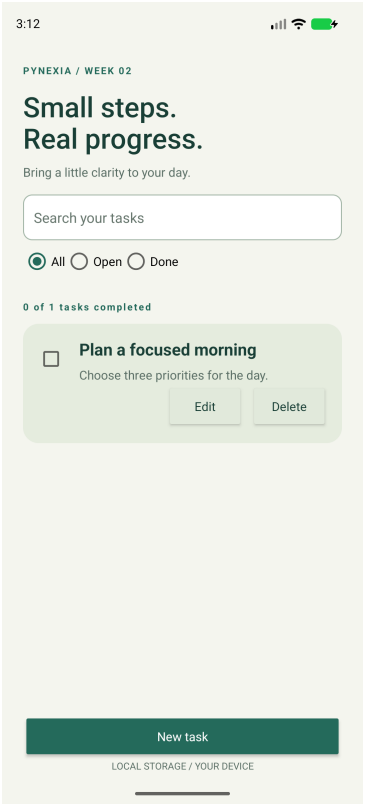
1. Tap New task, enter a title and optional details, then Save.
2. Mark the checkbox and confirm the title is crossed out.
3. Switch between Open and Done, then reopen the app to check persistence.
4. Edit, search, uncheck or delete a task. Cancelling deletion preserves it.

## Scope

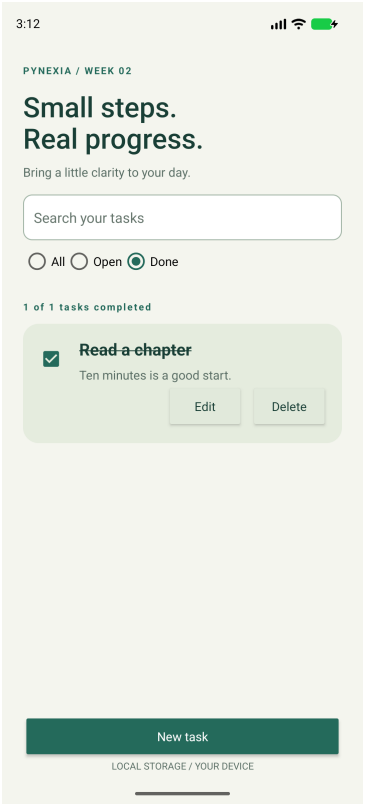
No reminders, due dates, notifications or cloud sync; these are outside the assignment. Clearing app data/uninstalling removes tasks. SQLite storage is not encrypted.

# Actual app screens

Captured from the tested native app on the emulator.




Saved task and details




Completed task and status filter

## Executed verification

4 JVM tests + 4 device tests passed. Lint: 0 errors, 18 warnings. No failures, errors or skipped tests.

Validation and search, SQLite CRUD/reopen, UI add/edit/delete/cancel, editor recreation, completion persistence and All/Open/Done filtering.

Pixel 8 emulator, Android 17 / API 37 (37.1 image), x86_64. No physical-device or API 26 test run. Lint warnings are retained without suppression. APKs are debug-signed evaluation builds.

## Reproduce

.\gradlew.bat :todo:assembleDebug :todo:testDebugUnitTest :todo:lintDebug

.\gradlew.bat :todo:connectedDebugAndroidTest

The second command needs a running emulator or connected device. Initial Gradle sync needs internet. See verification/ for executed test results and lint findings.

# Explanation & submission

ينظم التطبيق المهام اليومية مع الإضافة والتعديل والحذف وتحديد المهمة كمكتملة. يحتفظ بحالة الإنجاز محليا، ويتيح عرض جميع المهام أو المفتوحة أو المكتملة والبحث في محتواها. ترتبط الأزرار ومربعات الاختيار بالسجل الصحيح حتى عند تغير ترتيب القائمة.

## Repository URL

https://github.com/ENGEyad/pynexia-week2-todo

## Live Demo / Hosted URL

https://engeyad.github.io/pynexia-week2-todo/

A hosted project showcase with real screenshots and downloads, not an Android emulator.

## Attach File / Archive

To-Do-List-App.zip - source, Gradle wrapper, APK, screenshots, verification and this PDF. Below the 10 MB limit. APK is inside the ZIP because the earlier submission form does not list direct APK uploads.

## Notes / Implementation Remarks (1330 / 2000 characters)

Native Android To-Do List App implemented with Java and XML. Manage daily tasks with add, edit, delete and completion controls, a RecyclerView list, status filters and durable local SQLite storage.

Each task has a required title, optional details and a stable ID. SQLite stores all fields. A custom ListAdapter binds each item and its actions. A checkbox persists the done flag and adds a strike-through title. All/Open/Done filters and literal text search can be combined. The count reports completed versus total tasks. Invalid titles show field errors. Delete requires confirmation. Background database work avoids blocking the UI; ViewModel retains in-flight editor saves.

Run: open the source folder in Android Studio, sync Gradle, and run todo on Android 8.0/API 26 or later (compile/target SDK 35).

Validation: 4 JVM tests and 4 device tests passed on a Pixel 8 emulator, Android 17/API 37. Lint: zero errors; 18 warnings documented without suppression.

No reminders, due dates, notifications or cloud sync; these are outside the assignment. Clearing app data/uninstalling removes tasks. SQLite storage is not encrypted.

Hosted URL: project showcase with actual screenshots and downloads, not a browser emulator. The ZIP contains source, Gradle wrapper, debug APK, screenshots, verification and a bilingual PDF report.